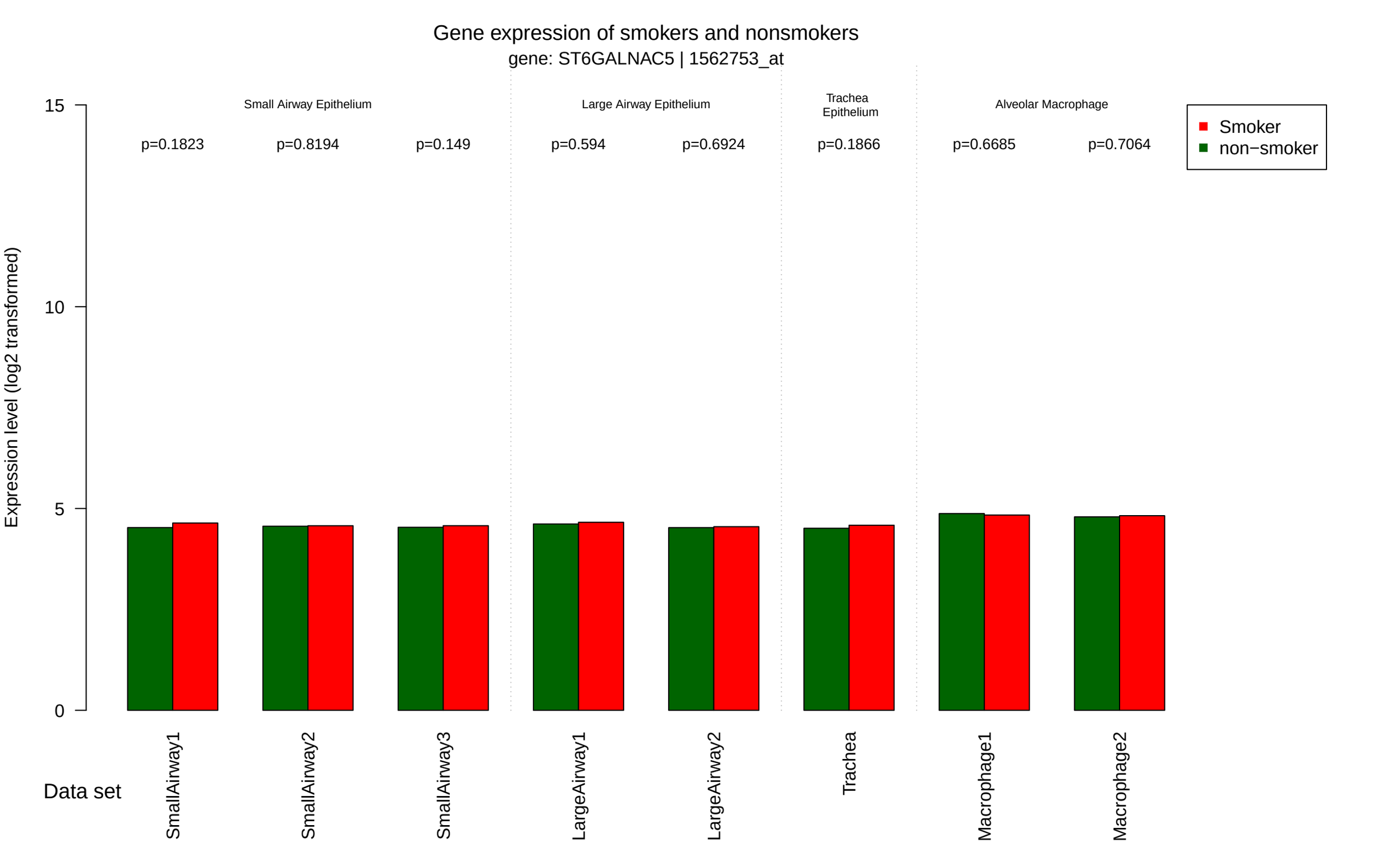
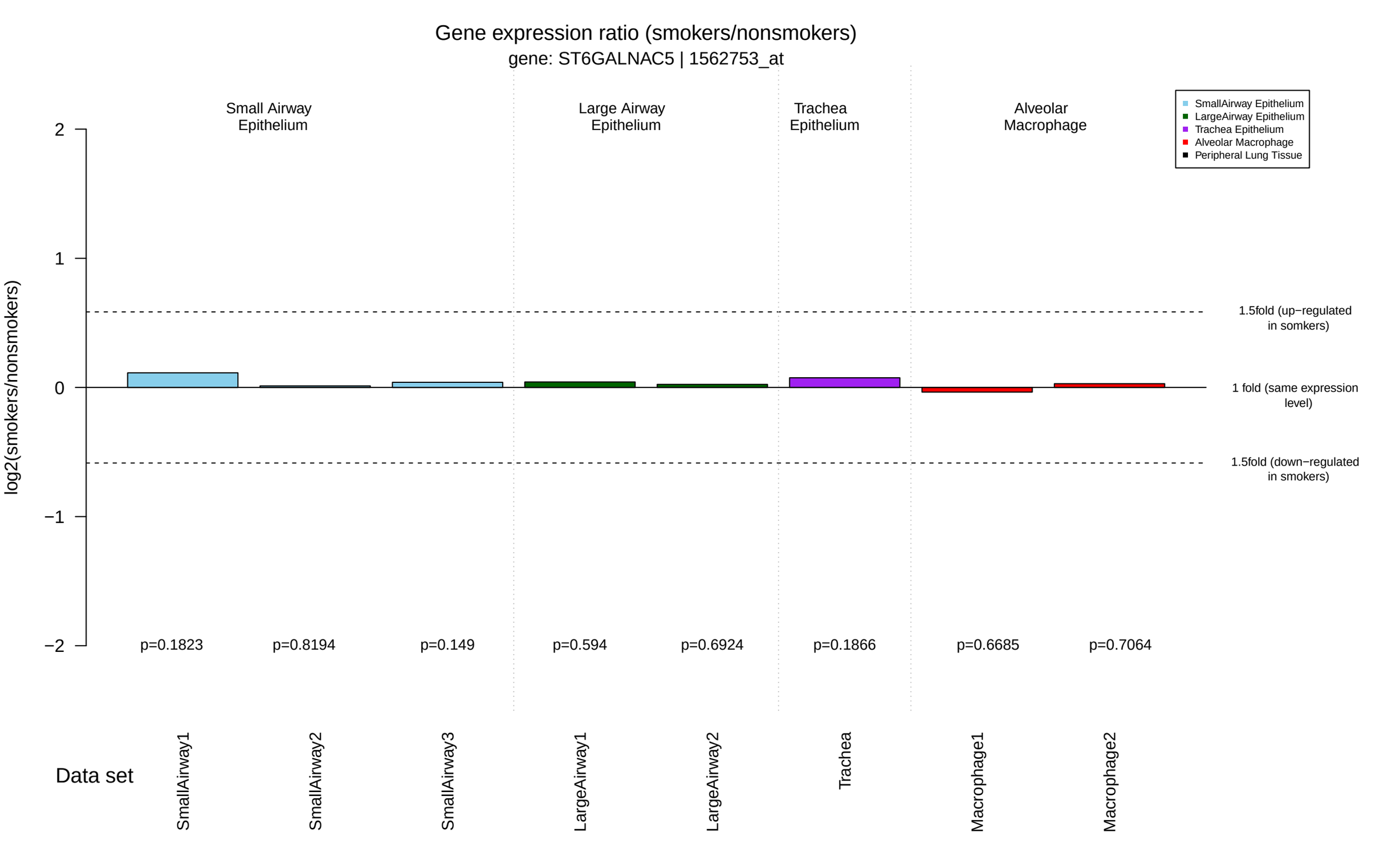
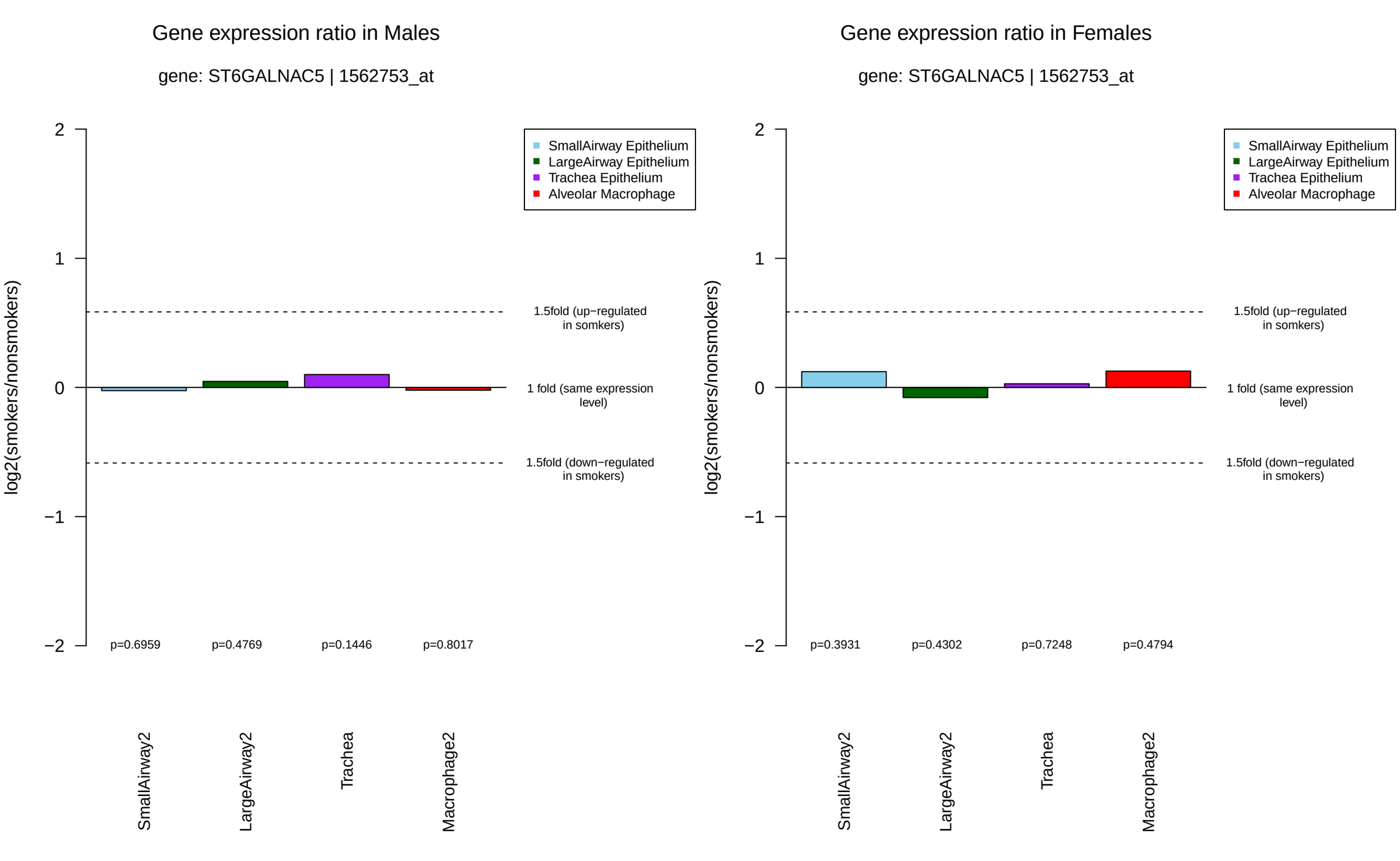
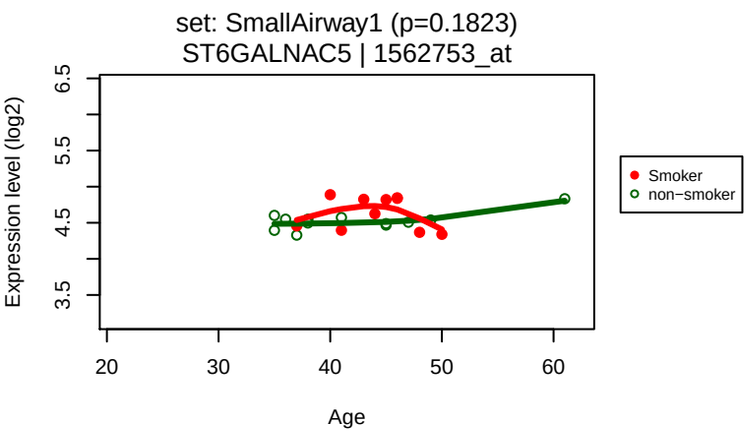
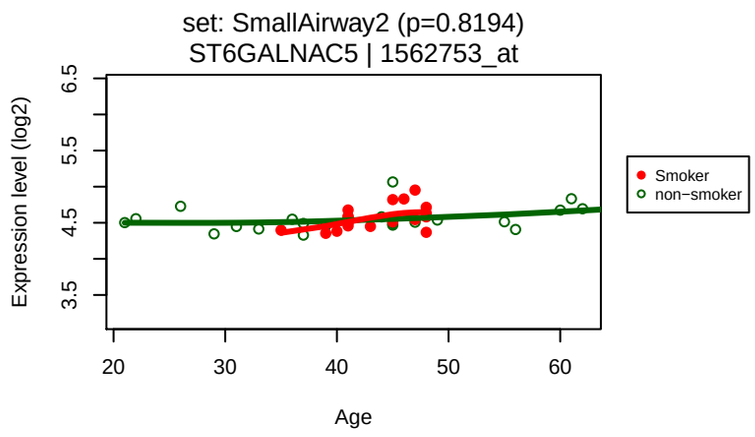
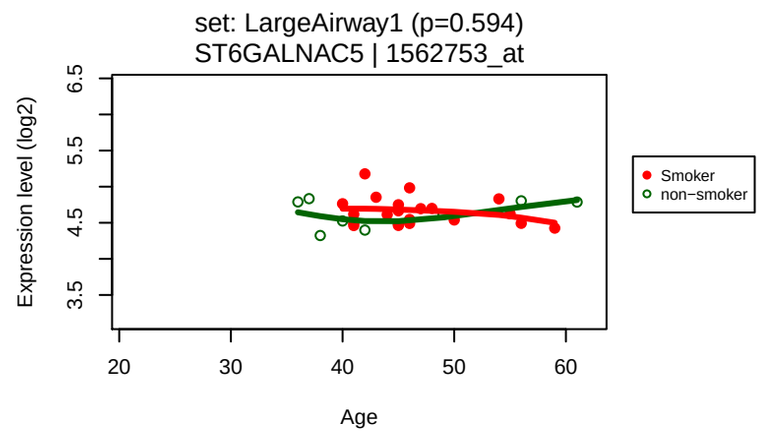

Gene expression of smokers and nonsmokers
gene: ST6GALNAC5 | 1562753_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1823
p=0.8194
p=0.149
p=0.594
p=0.6924
p=0.1866
p=0.6685
p=0.7064
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

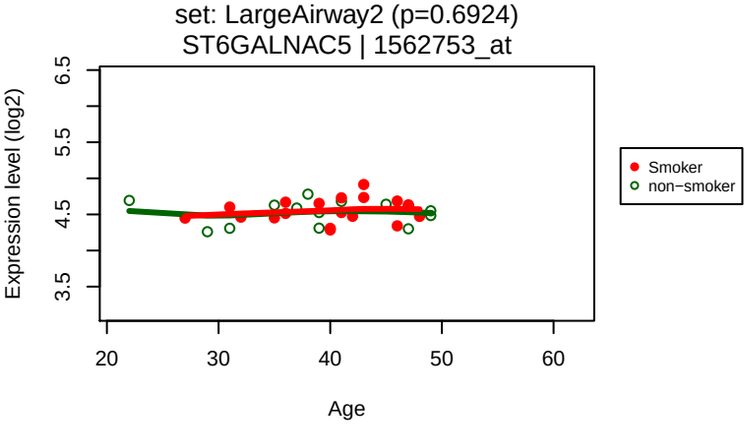

Gene expression ratio (smokers/nonsmokers)
gene: ST6GALNAC5 | 1562753_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1823
p=0.8194
p=0.149
p=0.594
p=0.6924
p=0.1866
p=0.6685
p=0.7064
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

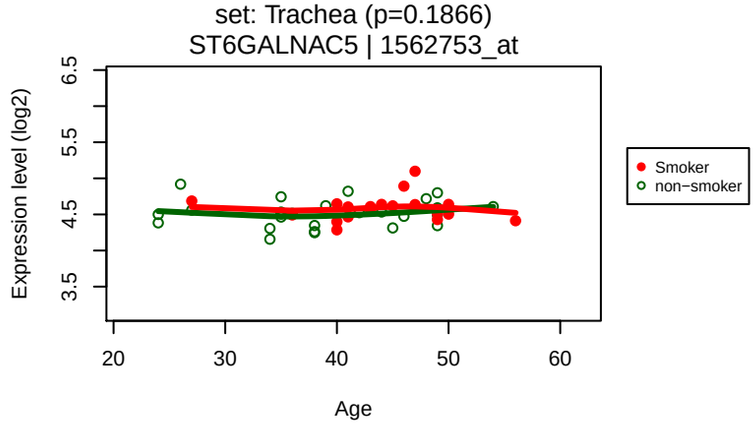

Gene expression ratio in Males
gene: ST6GALNAC5 | 1562753_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6959
p=0.4769
p=0.1446
p=0.8017
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST6GALNAC5 | 1562753_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3931
p=0.4302
p=0.7248
p=0.4794
SmallAirway2
LargeAirway2
Trachea
Macrophage2

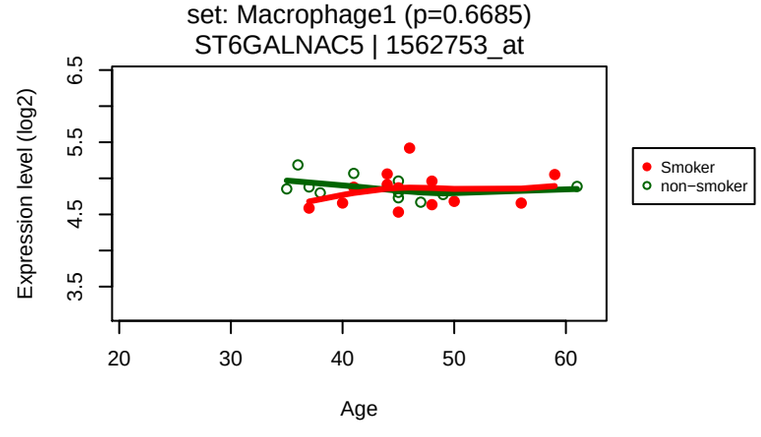

set: SmallAirway1 (p=0.1823)
ST6GALNAC5 | 1562753_at
Expression level (log2)
Age
Smoker
non−smoker

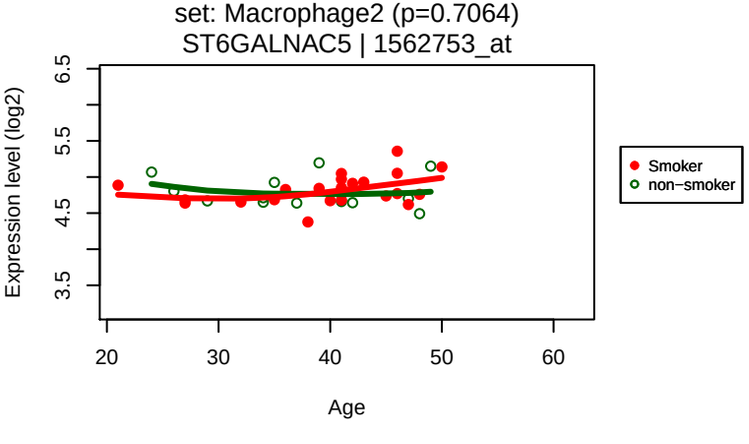

set: SmallAirway2 (p=0.8194)
ST6GALNAC5 | 1562753_at
Expression level (log2)
Age
Smoker
non−smoker

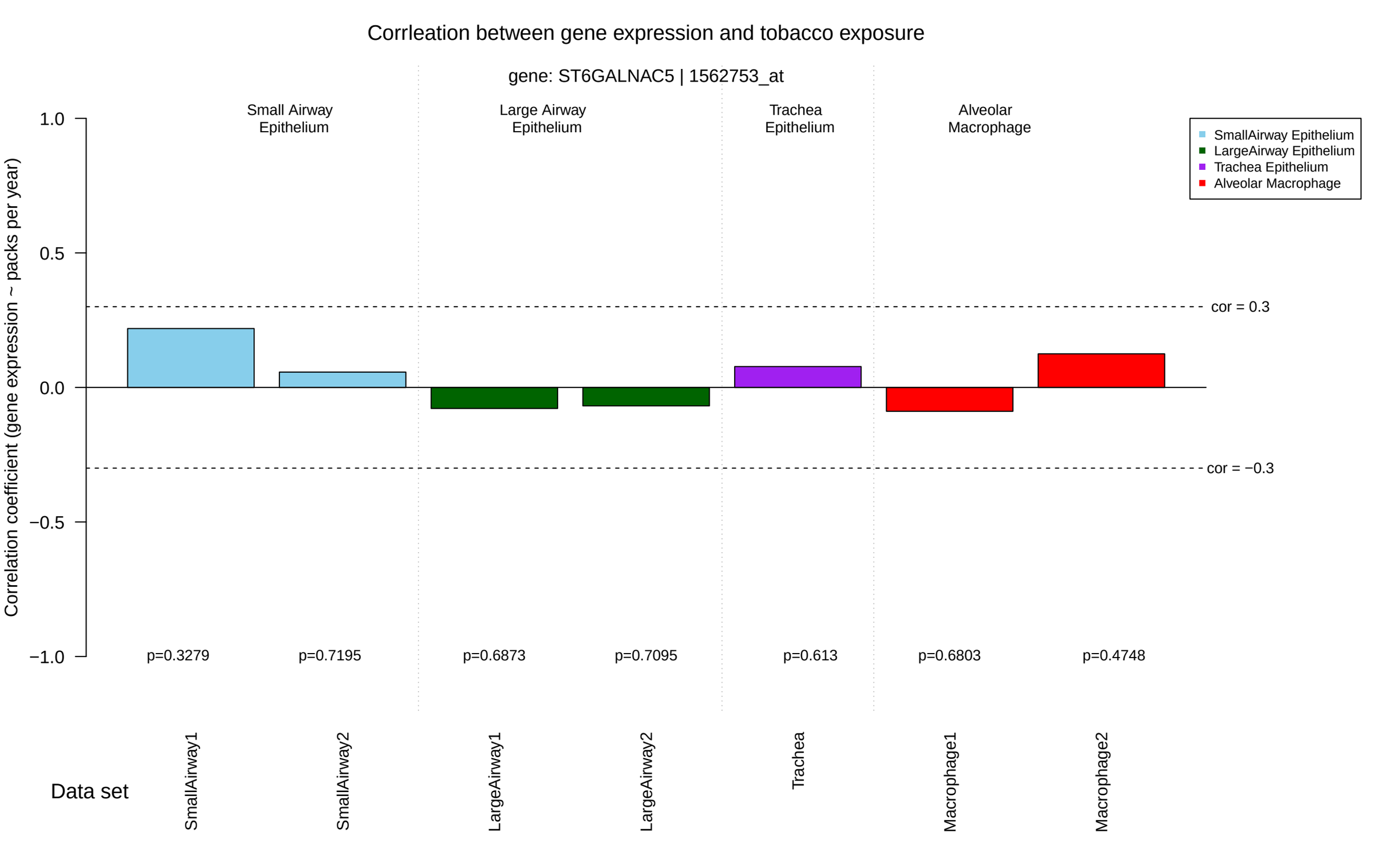

set: LargeAirway1 (p=0.594)
ST6GALNAC5 | 1562753_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6924)
ST6GALNAC5 | 1562753_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1866)
ST6GALNAC5 | 1562753_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6685)
ST6GALNAC5 | 1562753_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7064)
ST6GALNAC5 | 1562753_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST6GALNAC5 | 1562753_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3279
p=0.7195
p=0.6873
p=0.7095
p=0.613
p=0.6803
p=0.4748
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2